UNIVERSITÀ DEGLI STUDI DI PISA

CORSO DI STUDIO IN BIONICS ENGINEERING

ADVANCED SIGNAL PROCESSING - *Prova scritta preliminare del 15/06/2016*

EXERCISE

We observe N samples of a realization of the random process $X[n] = A + W[n]$, for n=0, 1,…, N-1. where A a constant signal of interest (SOI), modelled as an unknown deterministic parameter, and $\{W[n]; n = 0,1,\cdots, N-1\}$ are independent and identically distributed (IID) random variables modeling additive noise, which is independent of A. Each noise sample is Gaussian distributed, zero mean and with unknown variance σ_W^2. The signal to noise ratio (SNR) is as follows: $\gamma \triangleq A^2/\sigma_W^2$.

(1) Derive the autocorrelation function (ACF) $R_X[m] = E\{X[n]X[n+m]\}$ and the power spectral density (PSD) $S_X\left(e^{j2\pi f}\right) = FT\{R_X[m]\}$ of the observed signal $X[n]$ and plot them.

(2) Derive the joint maximum likelihood (ML) estimator of the unknown signal A and unknown noise power σ_W^2.

(3) Derive the bias and the mean square error (MSE) of the ML estimator of A, plot the MSE as a function of γ and find the minimum value of N which guarantees an MSE lower than 10^{-3} when the noise power is σ_W^2 =1.

(4) Derive the bias of the ML estimator of σ_W^2 and plot it as a function of the sample size N.

(5) Does it exist the linear Least Squares (LS) estimator of A? Does it exist the LS estimator of σ_W^2? [*Motivate the answers, but it is not require to find these estimators, if they exist*].

Solution:

(1) Derive the autocorrelation function (ACF) $R_X[m] = E\{X[n]X[n+m]\}$ and the power spectral density (PSD) $S_X\left(e^{j2\pi f}\right) = FT\{R_X[m]\}$ of the observed signal $X[n]$ and plot them.

$$R_X[m] = E\{X[n]X[n+m]\} = A^2 + \sigma_W^2 \delta[m]$$

$$S_X\left(e^{j2\pi f}\right) = FT\{R_X[m]\} = 2\pi A^2 \delta\left(e^{j2\pi f} - 1\right) + \sigma_W^2$$

(2) Derive the joint maximum likelihood (ML) estimator of the unknown signal *A* and unknown noise power σ_W^2.

$$\hat{A}_{ML} = \frac{1}{N}\sum_{n=0}^{N-1} X[n]$$

$$\hat{\sigma}_{W,ML}^2 = \frac{1}{N}\sum_{n=0}^{N-1}\left(X[n] - \hat{A}_{ML}\right)^2 = \frac{1}{N}\sum_{n=0}^{N-1}\left(X[n] - \frac{1}{N}\sum_{k=0}^{N-1} X[k]\right)^2$$

(3) Derive the bias and the mean square error (MSE) of the ML estimator of *A*, plot the MSE as a function of γ and find the minimum value of *N* which guarantees an MSE lower than 10^{-3} when the noise power is σ_W^2 =1.

$$b\left\{\hat{A}_{ML}\right\} = E\left\{A - \hat{A}_{ML}\right\} = 0$$

$$MSE\left\{\hat{A}_{ML}\right\} = \mathrm{var}\left\{\hat{A}_{ML}\right\} = \frac{\sigma_W^2}{N}$$

$$MSE\left\{\hat{A}_{ML}\right\} = \frac{\sigma_W^2}{N} < 10^{-3} \quad \rightarrow \quad N > 10^3 \sigma_W^2 = 10^3$$

(4) Derive the bias of the ML estimator of σ_W^2 and plot it as a function of the sample size *N*.

$$\hat{\sigma}^2_{W,ML} = \frac{1}{N}\sum_{n=0}^{N-1}\left(X[n] - \hat{A}_{ML}\right)^2 = \frac{1}{N}\sum_{n=0}^{N-1} X^2[n] - \left(\hat{A}_{ML}\right)^2$$

$$b\left\{\hat{\sigma}^2_{W,ML}\right\} = \sigma_W^2 - E\left\{\hat{\sigma}^2_{W,ML}\right\} = \sigma_W^2 - \left(A^2 + \sigma_W^2\right) + A^2 + \frac{\sigma_W^2}{N} = \frac{\sigma_W^2}{N}$$

(5) Does it exist the linear Least Squares (LS) estimator of *A*?

Since the noise is AWGN, we have: $\hat{A}_{LS} = \hat{A}_{ML} = \frac{1}{N}\sum_{n=0}^{N-1} X[n]$

Does it exist the LS estimator of σ_W^2 ?

$\hat{\sigma}^2_{W,LS}$ does not exist, since the signal is not a linear function of the unknown parameter σ_W^2